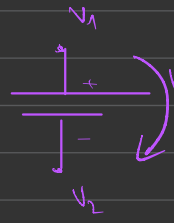
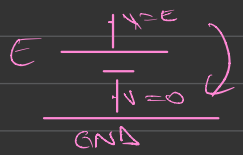
Seminar 1

Simboluri: 

sursă de tensiune (crează o diferență de potențial)

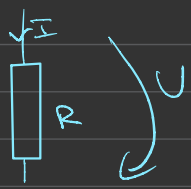
$U = V_1 - V_2 = E$



sursă conectată la ground

$V_1 - V = E \Rightarrow V_1 = E$

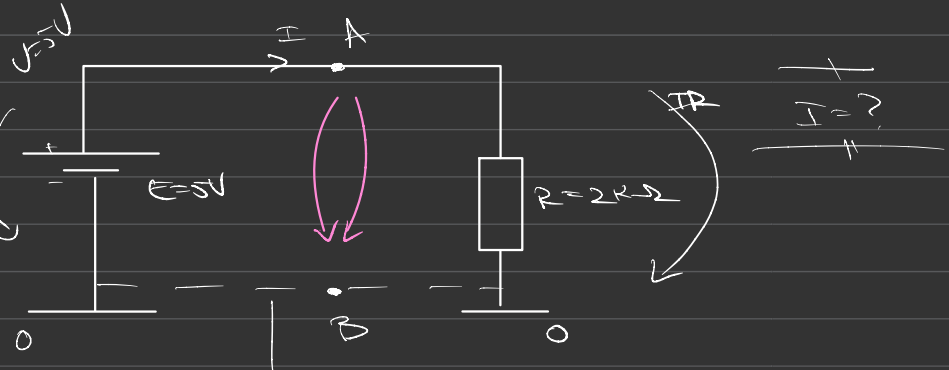
$V = 0$



legea lui Ohm $U = RI$

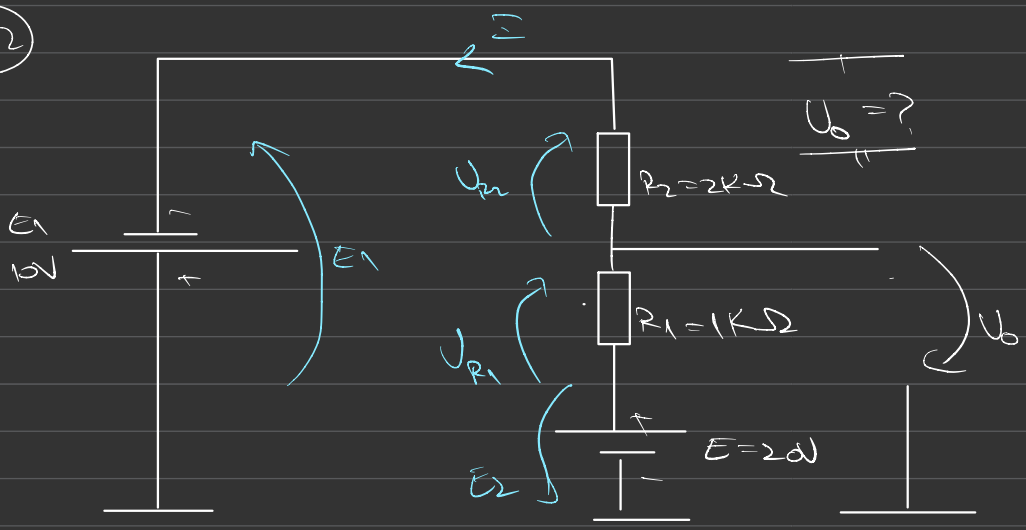
$I = \frac{U}{R}$

Ex 1



$\Rightarrow$ RVL KT: $E = IR$

Ex2



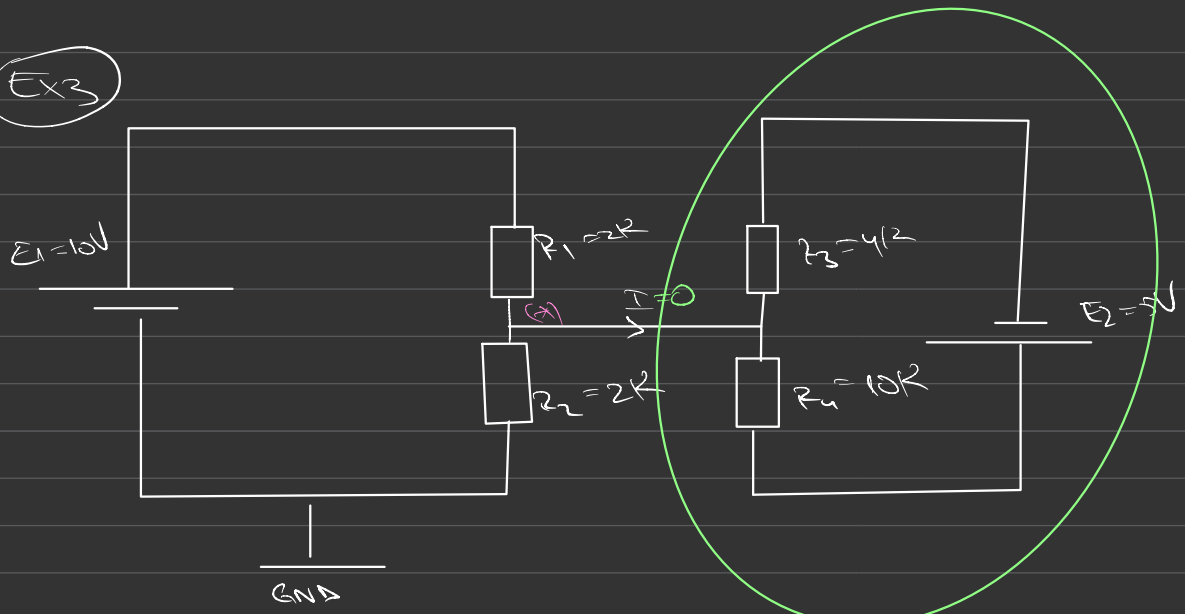
$$-E_1 = -IR_2 - IR_1 + E_2 \Rightarrow I = \frac{E_2 + E_1}{R_1 + R_2} = \frac{30V}{3K\Omega}$$

$$= 10mA$$

$$U_{R_1} + U_0 - E_2 = 0$$

$$U_0 = E_2 - IR_1 \Rightarrow U_0 = 20 - 10 \cdot 1 = 10V$$

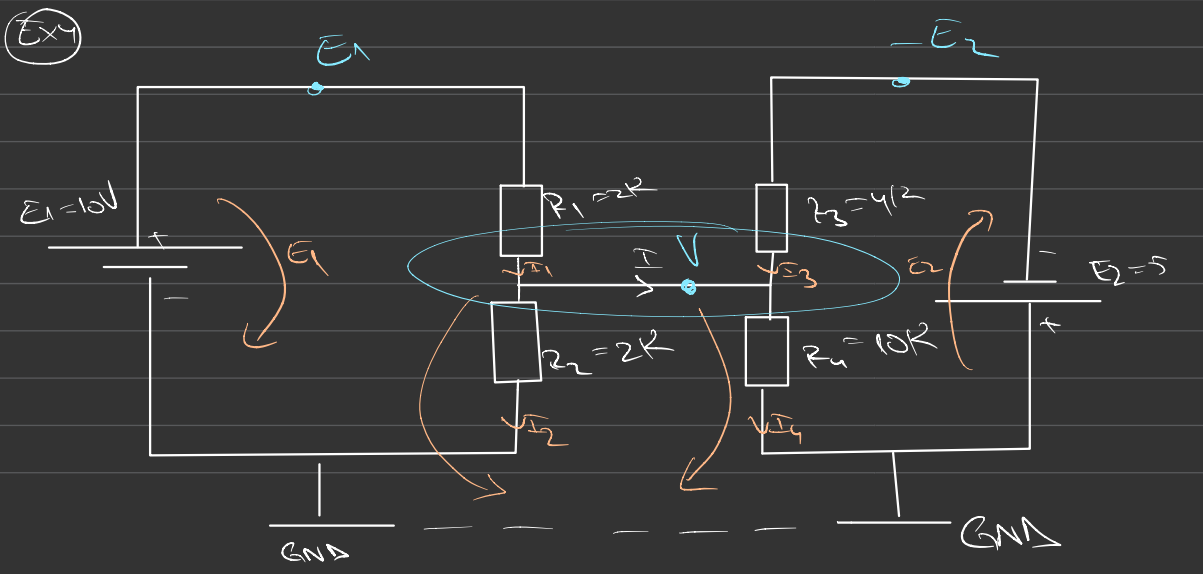
Ex3



KI: I=0

elettroni non sono più uniti se interrotti

Ex4



Avem o sursă de curent și la ea doi curenți.

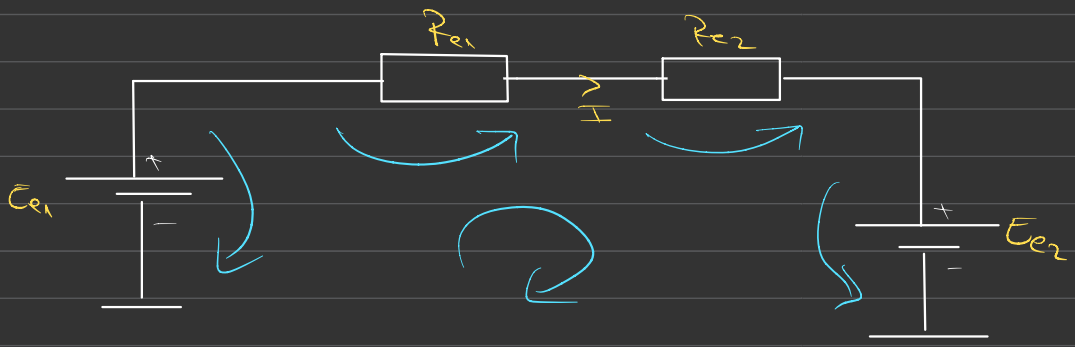
$$I = I_1 - I_2$$
$$E_2 + R_3 I_3 + R_4 I_4 = 0$$
$$I + I_3 = I_4$$
$$E_1 = R_1 I_1 + R_2 I_2$$
$$R_2 I_2 = I_4 R_4$$

Altă metodă (Potențialelor la noduri): $I_1 - I_2 + I_3 - I_4 = 0$

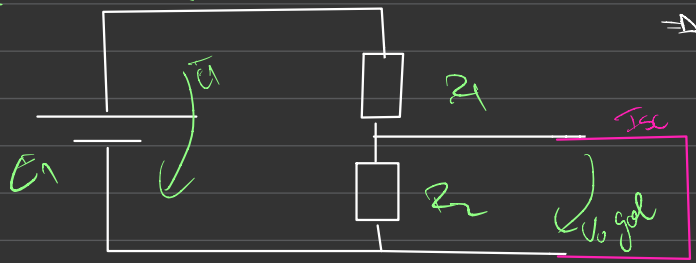
$$\frac{E_1 - V}{R_1} - \frac{V-0}{R_2} + \frac{-E_2 - V}{R_3} - \frac{V-0}{R_4} = 0$$

$$\left(\frac{\text{curent intrat} - \text{curent ieșit}}{\text{rezistență}} \right)$$

Echivalente Thevenin



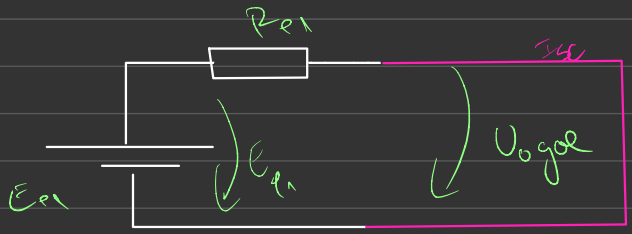
Divizor de tensiune



$\Rightarrow E_1 = I_{R_1} + I_{R_2}$

$I = \frac{E_1}{R_1+R_2}$

$U_{0gol} = \frac{E_1}{R_1+R_2} \cdot R_2$



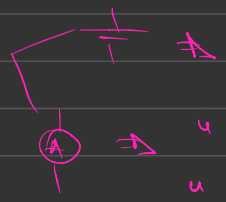
$\Rightarrow E_{e1} = U_{0gol}$

$U_{0gol} = E_{e1}$

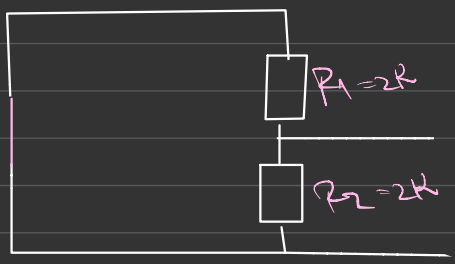
În general ⊙ $\boxed{E_e = U_{gol}}$ (Tensiune în gol. Fără consumator)

Rotim fata I.R. ca sa aflam rezistenta echivalenta

② $R_E = R$ CIRCUIT PASIVIZAT



Pasivizam circuitul anterior



$\Rightarrow R_{e1} = R_1 \| R_2 =$

$= \frac{R_1 R_2}{R_1 + R_2} = \frac{4}{4} = 1\ K\Omega$

$E_{e1} = \frac{E_1 R_2}{R_1 + R_2} = \frac{10 \cdot 2}{4} = 5V$

$\Rightarrow R_{e2} = R_3 \| R_4 = \frac{4 \cdot 10}{4+10} = \frac{40}{14} = 2{,}8\ K\Omega$

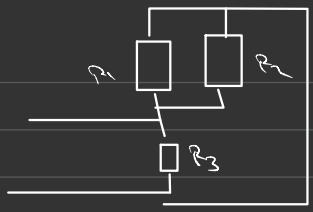
$E_{e2} = \frac{E_2 R_4}{R_3 + R_4} = 3{,}57\ V$

$-E_{e1} + IR_{e1} + IR_{e2} - E_{e2} = 0$

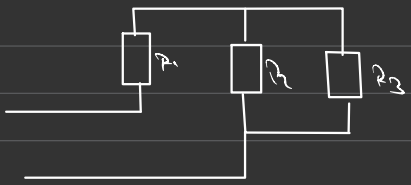
$I(R_{e1} + R_{e2}) = E_{e1} + E_{e2} \Rightarrow I = \frac{E_{e1} + E_{e2}}{R_{e1} + R_{e2}}$

$I = 2{,}247\ mA$

Es5



$R_e = R_1 \| R_2 \| R_3$



$R_e = R_1 + R_2 \| R_3$

